

Limitations of Quantum Sieving-Based Information-Set Decoding

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Introduction

The decoding problem underlies most of code-based cryptography

- Most efficient generic attacks: [information-set decoding](#) (ISD) algorithms
- Different variants exist
 - ▶ Including quantum ISD algorithms

New variant: ISD using a [‘code sieving’](#) subroutine [GJN23; DEEK24]

- Sieving-based ISD
- Promising asymptotic performance
- Sieving techniques are abundant in lattice-based cryptanalysis
 - ▶ Classical [BDGL16] and quantum [BCSS23] state-of-the-art for SVP

Introduction (cont.)

Natural questions:

- (1) Can we obtain **quantum speed-ups** for code sieving (similar to those observed in lattice sieving)?
- (2) How well do **quantum analogs of sieving-based ISD** perform?

Our work:

- (1) $\implies$ Yes!
- (2) $\implies$ Not so well...
- We tried to resolve (2), but **failed**. Hopefully an *insightful* failure!



Roadmap

- 1 Decoding Problem and ISD
- 2 Code Sieving and Sieving-Based ISD
- 3 Quantum Sieving-Based ISD and Its Limitations
- 4 Failed Attempts
- 5 Conclusion

Decoding Problem (DP)

An $[n, k]$ **binary linear code** is a k -dimensional subspace $\mathcal{C} \subseteq \mathbb{F}_2^n$. Elements of $\mathcal{C}$ are called **codewords**.

A fundamental problem in code-based cryptography:

Problem (Decoding Problem, $\text{DP}(n, k, w)$)

Given an $[n, k]$ binary linear code $\mathcal{C}$ and integer w , find $\mathbf{e} \in \mathcal{C}$ s.t. $|\mathbf{e}| = w$.

(Notation: $|\mathbf{x}|$ is the Hamming weight of $\mathbf{x} \in \mathbb{F}_2^n$.)

Some remarks:

- Given n, k , we can choose w such that $\exists$ unique solution on average.
 - ▶ **Unique decoding regime**
- In that regime, the decoding problem is **believed to be hard** for a random instance of the problem (i.e., when $\mathcal{C}$ is a **random code**).
 - ▶ All known algorithms run in time and memory exponential in n .

Algorithms for the Decoding Problem

Information-set decoding (ISD) algorithms:

- Most efficient generic attacks against the decoding problem
- All can be viewed as improvements of [Prange's algorithm](#) [Pra62]
- Formulated as a framework by Finiasz and Sendrier [FS09]

Information-Set Decoding (ISD)

Consider a $\text{DP}(n, k, w)$ -instance $\mathcal{C}$ with (unique) solution $\mathbf{e}$.
Thus, $\mathbf{e} \in \mathcal{C}$ and $|\mathbf{e}| = w$.

High-level idea (for parameters (n', w', N) to be optimized):

- Create a random $\text{DP}(n', k, w')$ -instance $\mathcal{C}'$ from $\mathcal{C}$.
 - ▶ A codeword in $\mathcal{C}'$ uniquely **lifts** to a codeword in $\mathcal{C}$.
 - ▶ **Hope**: One of sub-solutions lifts to the main solution $\mathbf{e}$.
- Uses an **oracle** $\mathcal{A}$ to find N solutions to this $\text{DP}(n', k, w')$ -instance.
 - ▶ This imposes an **upper bound** $N \leq \binom{n'}{w'} / 2^{n'-k}$.
- **Repeat** if none of the found sub-solutions yields a main solution.

Classical time complexity: $T_{\text{ISD}} = \tilde{\mathcal{O}}\left(\frac{T_{\mathcal{A}}}{p}\right)$

- $T_{\mathcal{A}}$: runtime of $\mathcal{A}$
- p : probability that one iteration succeeds

Quantum ISD and Its Complexity

Quantum variant:

- Suffices to repeat $\frac{1}{\sqrt{p}}$ times (amplitude amplification [BHMT02])
- Oracle $\mathcal{A}$ is allowed to be quantum
 - ▶ Runtime $T_{\mathcal{A}}^{(Q)}$

Quantum time complexity: $T_{\text{ISD}}^{(Q)} = \tilde{\mathcal{O}}\left(\frac{T_{\mathcal{A}}^{(Q)}}{\sqrt{p}}\right)$ instead of $T_{\text{ISD}} = \tilde{\mathcal{O}}\left(\frac{T_{\mathcal{A}}}{p}\right)$

Existing quantum ISD algorithms:

- Quantum Prange: [Ber10]
 - ▶ Prange's algorithm + amplitude amplification
- Improvements: [KT17; Kir18]
 - ▶ Improved runtime, but worse memory complexities than quantum Prange

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Sieving-Based ISD

Sieving-based ISD: ISD with oracle $\mathcal{A}$ instantiated with a ‘sieving algorithm’.

Recall that $\mathcal{A}$ aims to find N codewords of weight w' in an $[n', k]$ code $\mathcal{C}'$.

Oracle $\mathcal{A}$: Code Sieving [GJN23; DEEK24]

Starting from N random vectors of weight w' in $\mathbb{F}_2^{n'}$ (not necessarily in $\mathcal{C}'$), ‘sieve’ those vectors that are in $\mathcal{C}'$.

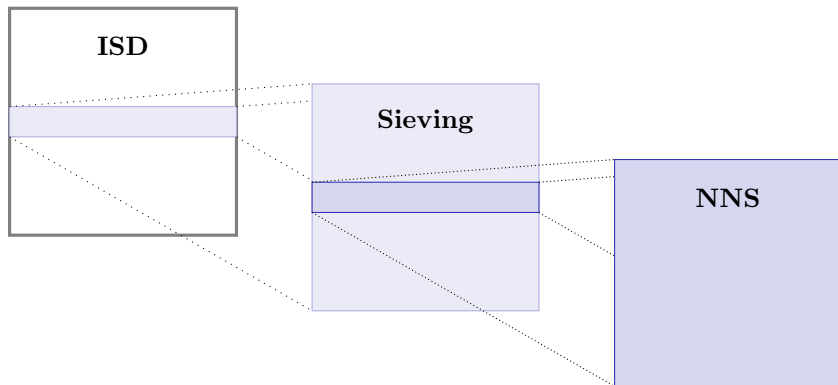
The sieving is performed **iteratively**:

- In iteration i :
 - ▶ **Given:** N vectors of weight w' satisfying $i - 1$ code constraints.
 - ▶ **Goal:** **Combine them** to obtain N new vectors of weight w' satisfying i code constraints. $\rightsquigarrow$ *Near-neighbor search (NNS)*
- After $n' - k$ iterations, we end up with N codewords in $\mathcal{C}'$ of weight w' .

This code-sieving approach imposes a **lower bound** on $\mathcal{A}$ ’s output size:

$$N \geq 4 \frac{\binom{n'}{w'}}{\binom{w'}{w'/2} \binom{n'-w'}{w'/2}}.$$

ISD, Code Sieving, and NNS



Credits: Simona Etinski

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Quantum Sieving-Based ISD

ISD algorithms using code sieving:

- *Sieving-based ISD*
 - ▶ Classical ISD where $\mathcal{A}$ is instantiated with a code-sieving algorithm
- *Quantum sieving-based ISD*
 - ▶ Quantum ISD where $\mathcal{A}$ is instantiated with a code-sieving algorithm
 - ▶ Recall: Code-sieving algorithm may be quantum!

Recall the ISD parameters (n', w', N) : $\mathcal{A}$ finds N solutions to $\text{DP}(n', k, w')$.

Our result:

- Consider any (n', w', N) that satisfy the **lower bound constraint** on N .
 - ▶ This is the case for existing code-sieving algorithms.
- **Then the runtime of quantum sieving-based ISD is at least that of quantum Prange.**¹

(This is in contrast to classical sieving-based ISD!)

¹Consequently, it performs worse than the quantum state-of-the-art [KT17; Kir18].

Limitations of Quantum Sieving-Based ISD

Claim 1

Consider a random instance of $\text{DP}(n, k, w)$ in the unique decoding regime. For all $n', w', N \in \mathbb{N}$ satisfying $4 \binom{n'}{w'} / \left(\binom{w'}{w'/2} \binom{n'-w'}{w'/2} \right) \leq N \leq \binom{n'}{w'} / 2^{n'-k}$, there is no $\mathcal{A}$ such that the resulting quantum ISD algorithm has a better runtime than quantum Prange.

Proof sketch:

- The expected runtime of quantum ISD is $T = T_{\mathcal{A}}^{(Q)} / \sqrt{p}$.
 - ▶ The probability p is of the form $p = N \cdot q$, where q is independent of N .
 - ▶ A trivial lower bound on $T_{\mathcal{A}}^{(Q)}$ is N .
- So $T \geq N / \sqrt{N \cdot q} = \sqrt{N/q}$.
- By the lower bound on N , we obtain $T \geq T_{n', w'}$ in terms of n', w' alone.
- We then numerically show that (for all allowed n', w') this lower bound $T_{n', w'}$ is at least the runtime of quantum Prange.

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Attempts to Overcome the Limitations

Bottleneck for quantum sieving-based ISD: the lower bound on N , i.e., on the output size of the sieving subroutine $\mathcal{A}$.

We present two failed attempts to circumvent this issue.

NB: We will omit constant factors in this high-level explanation.

Failed Attempt 1: From Pairs to Tuples

Let's zoom in on the source of the lower bound constraint on N .

- **Sieving algorithm:** In each iteration:
 - ▶ **Given:** List $\mathcal{L}$ of N weight- w' vectors
 - ▶ **Goal:** Find N pairs $(\mathbf{x}_1, \mathbf{x}_2) \in \mathcal{L}^2$ s.t. $|\mathbf{x}_1 + \mathbf{x}_2| = w'$. (NNS)
- The **lower bound on N** comes from guaranteeing that the expected number of solution pairs is $\geq N$.
 - ▶ Need $N^2 p^{(2)} \geq N$, where $p^{(2)} = \binom{w'}{w'/2} \binom{n'-w'}{w'/2} / \binom{n'}{w'}$.

Idea from lattice sieving: Use t -tuples to reduce the lower bound on N .

For some $t \geq 2$:

- **t -Sieving algorithm:** In each iteration:
 - ▶ **Given:** List $\mathcal{L}$ of N weight- w' vectors
 - ▶ **Goal:** Find N tuples $(\mathbf{x}_1, \dots, \mathbf{x}_t) \in \mathcal{L}^t$ s.t. $|\mathbf{x}_1 + \dots + \mathbf{x}_t| = w'$.
- This may result in a *lower* lower bound on the output size N .
 - ▶ Need $N^t p^{(t)} \geq N$, where $p^{(t)}$ is the probability that a random t -tuple forms a solution tuple.

Failed Attempt 1: From Pairs to Tuples

But... this fails.

Corollary 1 of Claim 1

Let $t \geq 2$. Assume that the runtime of any t -sieving algorithm (of the aforementioned kind) is lower bounded by the optimal runtime of 2-sieving.

Then, for all quantum ISD algorithms with a t -sieving algorithm as oracle $\mathcal{A}$, the expected runtime is no better than that of quantum Prange.

Failed Attempt 2: Varying List Sizes

The lower bound on N comes from **maintaining the same list size N** throughout each iteration of the sieving subroutine.

- Given N vectors, aims to find N new vectors for the next iteration.
- *Natural question:* Can we overcome the issue by **varying** the list sizes?

Consider (for some n', w') a sieving subroutine of the following form:

1. Sample a list $\mathcal{L}_0$ of N_0 vectors of weight w' .
2. In iteration $i = 1, \dots, n' - k$:
 - a. Find N_i pairs $(\mathbf{x}_1, \mathbf{x}_2) \in \mathcal{L}_{i-1}^2$ s.t. $|\mathbf{x}_1 + \mathbf{x}_2| = w'$.²
 - b. Let $\mathcal{L}_i$ be the resulting list.

Corollary 2 of Claim 1

For all possible choices of N_i , the expected runtime of quantum ISD with the above algorithm as oracle $\mathcal{A}$ is no better than that of quantum Prange.

²And discard those not satisfying the i -th code constraint.

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Conclusion

We studied a **natural quantum analog of sieving-based ISD**.

- Quantum speed-ups **are possible** for code sieving (and for NNS in $\mathbb{F}_2^n$) *when viewed as a problem in itself*.
- But... the resulting **quantum sieving-based ISD algorithm** does *not* yield runtime comparable to the quantum state-of-the-art.
- Main **bottleneck**: Lower-bound constraint on N
 - ▶ Imposed by (existing approaches for) code sieving
- Two natural attempts to resolve:
 - (1) Tuple sieving
 - (2) Varying list sizes
- Both **failed**.

This work hopefully provides useful insight into some intrinsic limitations of sieving-based ISD **from a quantum perspective**.

Thank you!



(Scan or click on the QR code for details about our work. See the next slide for references.)

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